

Problem 29.8: Integrating Out Right-Handed Neutrinos

Source: M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Problem 29.8.

Show that when you integrate out the right-handed neutrinos in Eq. (29.63), a dimension-5 operator like that in Eq. (29.65) results. What is the exact relationship between M_{ij} and $\widetilde{M}_{ij}$?

Solution. The relevant Lagrangian for the right-handed neutrino is

$$\mathcal{L} = i\bar{\nu}_R^i(\not{\partial} - ig'Y_\nu\not{B})\nu_R^i - Y_{ij}^\nu\bar{L}^i\tilde{H}\nu_R^j - iM_{ij}(\nu_R^i)^c\nu_R^j, \quad (1.1)$$

where the hypercharge Y_ν vanishes. The equation of motion is given by

$$-i(\partial_\mu\nu_R^i)^\dagger\sigma^\mu - Y_{ji}^\nu(\bar{L}^j\tilde{H}) - 2iM_{ji}(\nu_R^j)^T\sigma_2 = 0. \quad (1.2)$$

At low energy compared to M_{ij} , the kinetic term of the right-handed neutrino can be neglected, and we find

$$-iM_{ij}(\nu_R^j)^T\sigma_2 = -\frac{1}{2}Y_{ji}^\nu(\bar{L}^j\tilde{H}), \quad M_{ij}\nu_R^j = -\frac{1}{2}i\sigma_2Y_{ij}^\nu(\bar{L}^j\tilde{H})^T, \quad (1.3)$$

where we have used $M_{ij} = M_{ji}$. Therefore, we find

$$-iM_{ij}(\nu_R^i)^c\nu_R^j = \frac{i}{4}Y_{ij}^\nu M_{jk}^{-1}Y_{kl}^\nu(\bar{L}^i\tilde{H})\sigma_2(\bar{L}^l\tilde{H})^T, \quad (1.4)$$

which gives

$$\boxed{\widetilde{M}_{il} = -\frac{i}{4}Y_{ij}^\nu M_{jk}^{-1}Y_{kl}^\nu} \quad (1.5)$$